Computer Graphics Notes

Chapter 1 : Introduction

Computer Graphics

- Computer graphics is the science and art of communicating visually via a computer's display and its interaction devices. The visual aspect of the communication is usually in the computer-to-human direction, with the human-to-computer direction being mediated by devices like the mouse, keyboard, joystick, game controller, or touch-sensitive overlay.
- Computer graphics is a cross-disciplinary field in which physics, mathematics, human perception, human-computer interaction, engineering, graphic design, and art all play important roles.
- In the field of computer graphics, the word "model" can refer to a geometric model or a mathematical model. A **geometric model** is a model of something we plan to have appear in a picture: We make a model of a car, or a house, or an armadillo. The geometric model is enhanced with various other attributes that describe the color or texture or reflectance of the materials involved in the model. Starting from nothing and creating such a model is called **modeling**, and the geometric-plus-other-information description that is the result is called a **model.**
- A **mathematical** is a model of a physical or computational process.
- **Computer Graphics** are defined as any sketch or a drawing or a special network that pictorially represents some meaningful information.
- **Computer Graphics** is used where a set of images needs to be manipulated or the creation of the image in the form of pixels and is drawn on the computer.
- **Computer Graphics** can be used in digital photography, film, entertainment, electronic gadgets, and all other core technologies which are required. It is a vast subject and area in the field of computer science.
- **Computer Graphics** can be used in UI design, rendering, geometric objects, animation, and many more. In most areas, computer graphics is an abbreviation of **CG**.
- There are several tools used for the implementation of Computer Graphics. The basic is the <graphics.h> header file in Turbo-C, Unity for advanced and even OpenGL can be used for its Implementation.

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- It was invented in 1960 by great researchers Verne Hudson and William Fetter from Boeing.

Computer Graphics refers to several things:

- The manipulation and the representation of the image or the data in a graphical manner.
- Various technology is required for the creation and manipulation.
- Digital synthesis and its manipulation.

Types of Computer Graphics

- **Raster Graphics:** In raster, graphics pixels are used for an image to be drawn. It is also known as a bitmap image in which a sequence of images is into smaller pixels. Basically, a bitmap indicates a large number of pixels together.
- **Vector Graphics:** In vector graphics, mathematical formulae are used to draw different types of shapes, lines, objects, and so on.

Elements

The **seven** basic elements of graphic design are line, shape, color, texture, type, space and image.

1. **Lines** are always more than just points that are strung together. Depending on their form, weight, length and context, lines can help organize information, define shapes, imply movement, and convey emotions.

   When it comes to selecting the appropriate lines for projects, designers have plenty of options. **Lines** can:
   - …be **horizontal**, vertical or diagonal.
   - …be **straight**, curved or freeform.
   - …**zigzag** or create other patterns.
   - …be **solid**, broken or implied.
2. **Shapes** are best understood as areas, forms or figures contained by a boundary or closed outline. There are **two** types of shapes that every graphic designer should understand: **geometric** and **organic** (or "free-flowing").
   - **Geometric shapes** can include either two-dimensional or three-dimensional forms. Geometric shapes can include

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triangles, pyramids, squares, cubes, rectangles, pentagons, hexagons, octagons, decagons, circles, ellipses and spheres.

- **Organic shapes** are far less uniform, proportional and well-defined. They can be symmetrical or asymmetrical. They might include natural shapes, such as leaves, crystals, and vines, or abstract shapes, such as blobs and squiggles.

3. **Color** can be a useful tool for communicating a mood or provoking an emotional response from your viewer. Color theory and the color wheel provide a practical guide for graphic designers who want to select a single color or combine multiple colors in a harmonious- or intentionally discordant-way.
    - **Primary colors (red, yellow and blue)** are defined as the pure-pigment colors from which all others are made. There is no way to mix any other color to get red, yellow or blue. But mix them together, and you create all kinds of shades.
    - **Secondary colors (violet, green and orange)** are the immediate results of mixing two primary colors: Red and yellow make orange; blue and red make purple; and yellow and blue make green.
    - **Tertiary colors (red-orange, yellow-orange, yellow-green, blue-green, blue-violet and red-violet)** are the six colors that result from mixing a primary color and a secondary color.
4. **Texture** is the feel of a surface—furry, smooth, rough, soft, gooey or glossy. Most graphic designers must visually convey texture by using illusions to suggest how their work might feel if viewers could touch it.
5. **Type :** Whether you're choosing a font or creating your own typography for a graphic design project, it's important to make sure the type you use is legible and appropriate for your subject. Type affects the overall mood of a design, so consider whether your letters should be print or script, and whether they should have angles that are sharp or rounded.
6. **Space :** Spacing is a vital part of any designer's toolkit. It can give a design breathing room, increase its visual impact, balance out heavier visual elements, and emphasize images or messages that viewers should remember. Without enough space, a design can risk becoming too visually cluttered for your audience to understand.
7. **Image :** Whether graphic designers use photographs or illustrations, they rely on images to grab the audience's attention and express specific

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messages. An image works on multiple levels simultaneously: It provides context for a designer's communication, adds necessary drama or action, and creates an overall mood.

Applications

- **Computer Graphics are used for** an **aided design for engineering and architectural system-** These are used in electrical automobiles, electro-mechanical, mechanical, electronic devices. For example gears and bolts.
- **Computer Art –** MS Paint.
- **Presentation Graphics –** It is used to summarize financial statistical scientific or economic data. For example- Bar chart, Line chart.
- **Entertainment-** It is used in motion pictures, music videos, television gaming.
- **Education and training-** It is used to understand the operations of complex systems. It is also used for specialized system such for framing for captains, pilots and so on.
- **Visualization-** To study trends and patterns. For example- Analyzing satellite photo of earth.

Cathode Ray Tube (CRT):

CRT stands for Cathode Ray Tube. CRT is a technology used in traditional computer monitors and televisions. The image on CRT display is created by firing electrons from the back of the tube of phosphorus located towards the front of the screen.

Once the electron heats the phosphorus, they light up, and they are projected on a screen. The color you view on the screen is produced by a blend of red, blue and green light.

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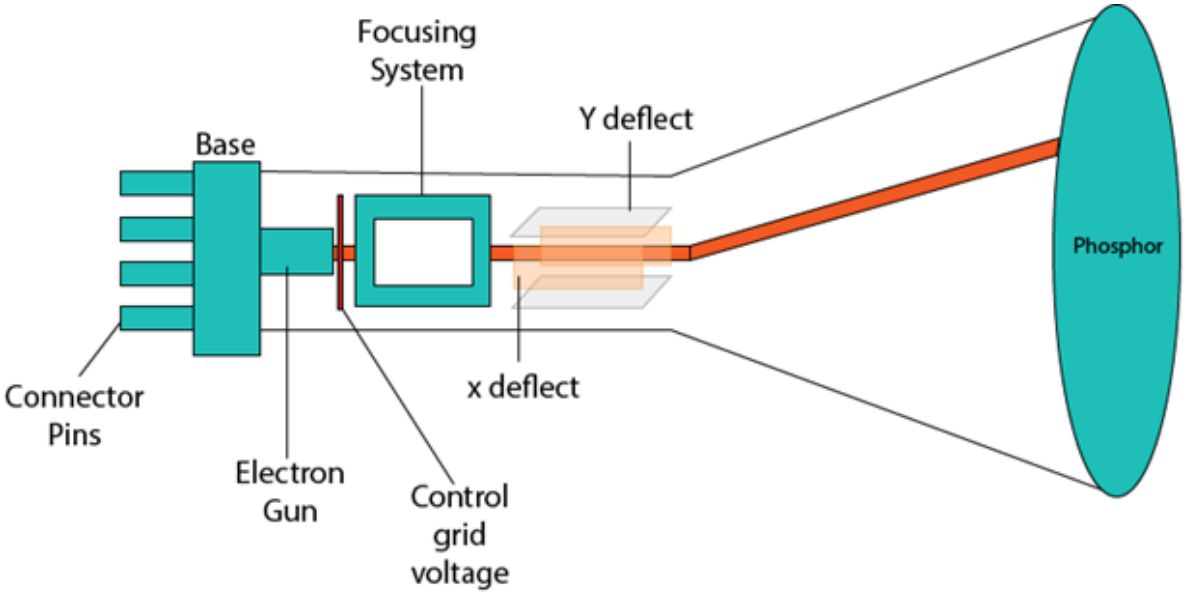

Cathode Ray Tube

Components of CRT:

Main Components of CRT are:

1. **Electron Gun:** Electron gun consisting of a series of elements, primarily a heating filament (heater) and a cathode. The electron gun creates a source of electrons which are focused into a narrow beam directed at the face of the CRT.

2. **Control Electrode:** It is used to turn the electron beam on and off.

3. **Focusing system:** It is used to create a clear picture by focusing the electrons into a narrow beam.

4. **Deflection Yoke:** It is used to control the direction of the electron beam. It creates an electric or magnetic field which will bend the electron beam as it passes through the area. In a conventional CRT, the yoke is linked to a sweep or scan generator. The deflection yoke which is connected to the sweep generator creates a fluctuating electric or magnetic potential.

5. **Phosphorus-coated screen:** The inside front surface of every CRT is coated with phosphors. Phosphors glow when a high-energy electron beam hits them. Phosphorescence is the term used to characterize the light given off by a phosphor after it has been exposed to an electron beam.

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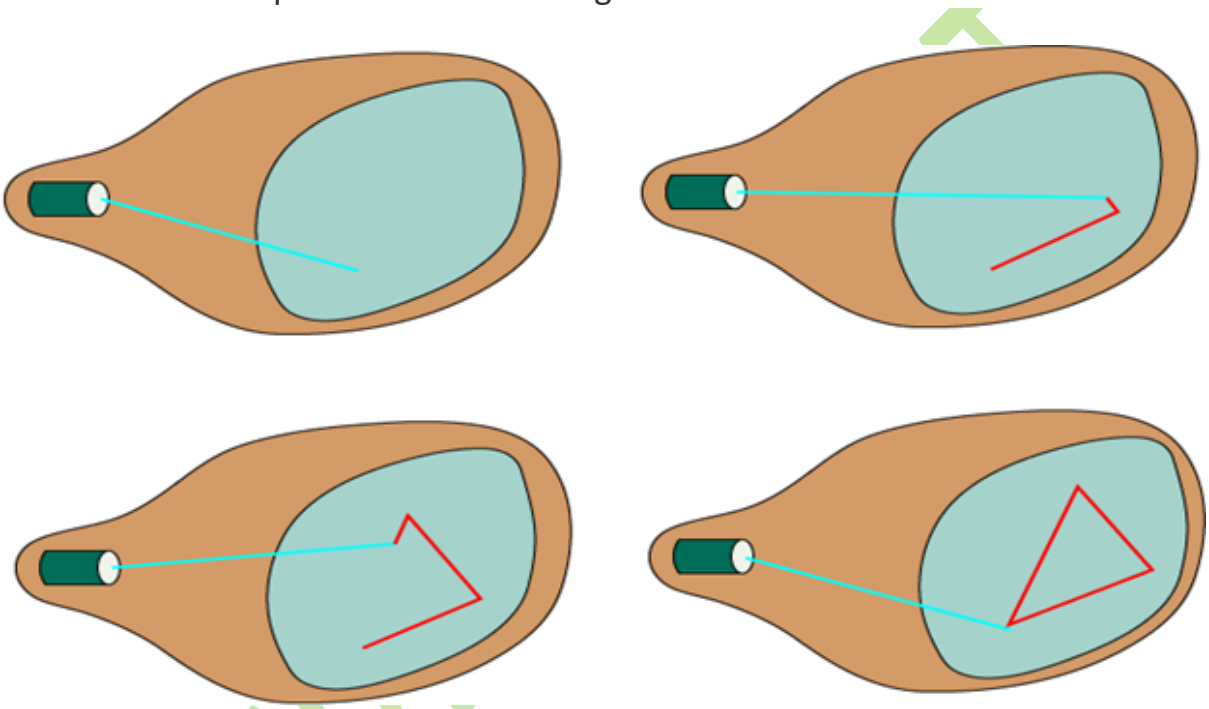
Random Scan Display:

Random Scan System uses an electron beam which operates like a pencil to create a line image on the CRT screen. The picture is constructed out of a sequence of straight-line segments. Each line segment is drawn on the screen by directing the beam to move from one point on the screen to the next, where its x & y coordinates define each point. After drawing the picture. The system cycles back to the first line and design all the lines of the image 30 to 60 time each second. The process is shown in fig:



Random-scan monitors are also known as vector displays or stroke-writing displays or calligraphic displays.

Advantages:

1. A CRT has the electron beam directed only to the parts of the screen where an image is to be drawn.
2. Produce smooth line drawings.
3. High Resolution

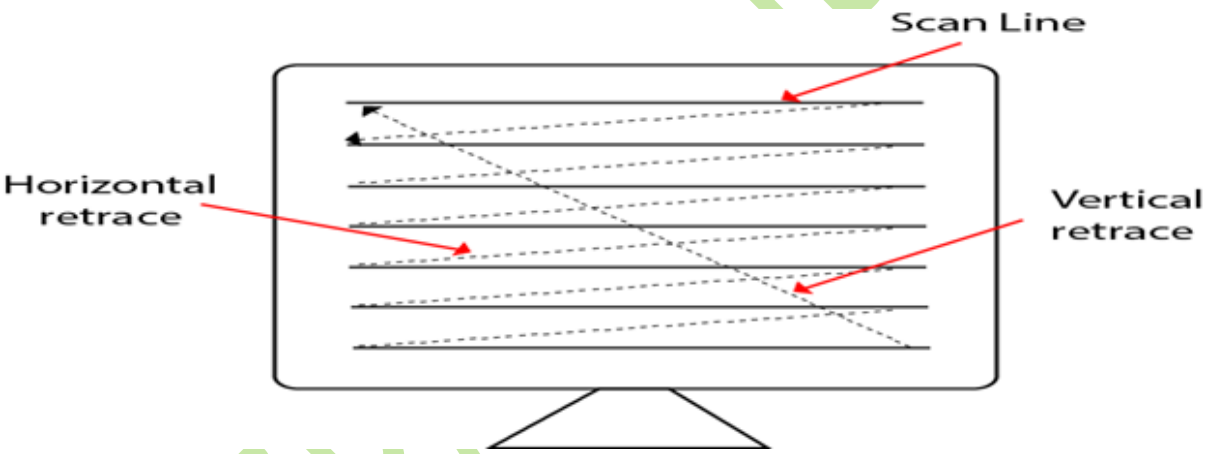
Disadvantages:

1. Random-Scan monitors cannot display realistic shades scenes.

Raster Scan Display:

A Raster Scan Display is based on intensity control of pixels in the form of a rectangular box called Raster on the screen. Information of on and off pixels is stored in refresh buffer or Frame buffer. Televisions in our house are based on Raster Scan Method. The raster scan system can store information of each pixel position, so it is suitable for realistic display of objects. Raster Scan provides a refresh rate of 60 to 80 frames per second.

Frame Buffer is also known as Raster or bit map. In Frame Buffer the positions are called picture elements or pixels. Beam refreshing is of two types. First is horizontal retracing and second is vertical retracing. When the beam starts from the top left corner and reaches the bottom right scale, it will again return to the top left side called at vertical retrace. Then it will again more horizontally from top to bottom call as horizontal retracing shown in fig:



Types of Scanning or travelling of beam in Raster Scan

1. Interlaced Scanning
2. Non-Interlaced Scanning

In Interlaced scanning, each horizontal line of the screen is traced from top to bottom. Due to which fading of display of object may occur. This problem can be solved by Non-Interlaced scanning. In this first of all odd numbered lines are traced or visited by an electron beam, then in the next circle, even number of lines are located.

For non-interlaced display refresh rate of 30 frames per second used. But it gives flickers. For interlaced display refresh rate of 60 frames per second is used.

Advantages:

1. Realistic image
2. Million Different colors to be generated
3. Shadow Scenes are possible.

Disadvantages:

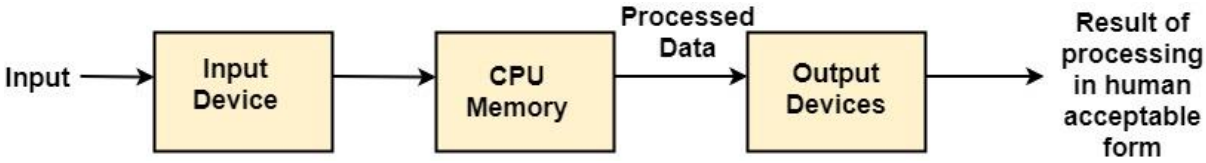
1. Low Resolution
2. Expensive

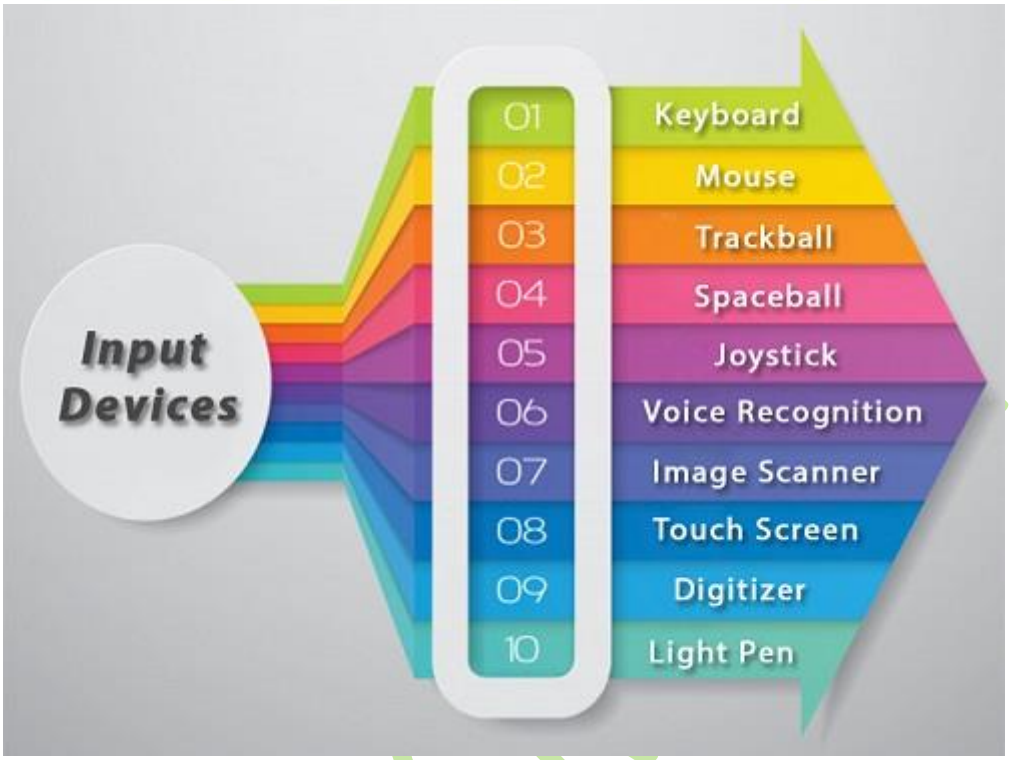
Differentiate between Random and Raster Scan Display:

Random Scan	Raster Scan
1. It has high Resolution	1. Its resolution is low.
2. It is more expensive	2. It is less expensive
3. Any modification if needed is easy	3.Modification is tough
4. Solid pattern is tough to fill	4.Solid pattern is easy to fill
5. Refresh rate depends or resolution	5. Refresh rate does not depend on the picture.
6. Only screen with view on an area is displayed.	6. Whole screen is scanned.
7. Beam Penetration technology come under it.	7. Shadow mark technology came under this.
8. It does not use interlacing method.	8. It uses interlacing
9. It is restricted to line drawing applications	9. It is suitable for realistic display.

Input Devices

The Input Devices are the hardware that is used to transfer transfers input to the computer. The data can be in the form of text, graphics, sound, and text. Output device display data from the memory of the computer. Output can be text, numeric data, line, polygon, and other objects.

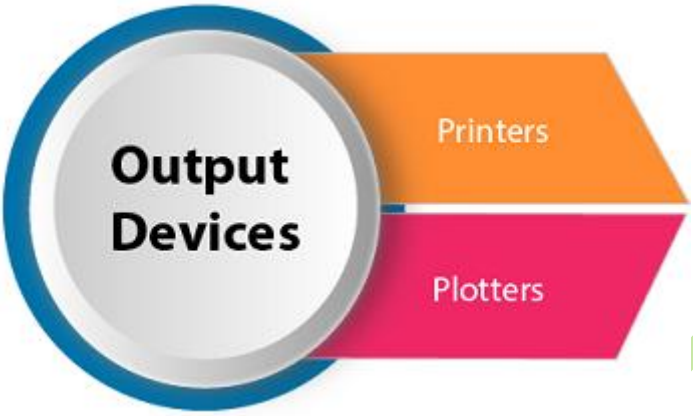




These Devices include:

1. Keyboard
2. Mouse
3. Trackball
4. Spaceball
5. Joystick
6. Light Pen
7. Digitizer
8. Touch Panels
9. Voice Recognition
10. Image Scanner

Output Devices



It is an electromechanical device, which accepts data from a computer and translates them into form understand by users.

Following are Output Devices:

1. Printers
2. Plotters

DDA Algorithm

DDA stands for Digital Differential Analyzer. It is an incremental method of scan conversion of line. In this method calculation is performed at each step but by using results of previous steps.

Suppose at step i, the pixels is (x_i,y_i)

The line of equation for step i

$y_i=mx_{i+b}$.....equation 1

Next value will be

$y_{i+1}=m_{xi+1}+b$.....equation 2

$m = \frac{\Delta y}{\Delta x}$

$y_{i+1}-y_i=\Delta y$.....equation 3

$y_{i+1}-x_i=\Delta x$.....equation 4

$y_{i+1}=y_i+\Delta y$

$\Delta y=m\Delta x$

$y_{i+1}=y_i+m\Delta x$

$\Delta x=\Delta y/m$

$x_{i+1}=x_i+\Delta x$
$x_{i+1}=x_i+\Delta y/m$

Case1: When $|M|<1$ then (assume that $x_1<x_2$)

$x= x_1, y=y_1$ set $\Delta x=1$
$y_{i+1}=y_{1+m}$, $x=x+1$
Until $x = x_2$

Case2: When $|M|<1$ then (assume that $y_1<y_2$)

$x= x_1, y=y_1$ set $\Delta y=1$

$x_{i+1}=\frac{1}{m}$, $y=y+1$

Until $y \rightarrow y_2$

Advantage:

1. It is a faster method than method of using direct use of line equation.
2. This method does not use multiplication theorem.
3. It allows us to detect the change in the value of x and y ,so plotting of same point twice is not possible.
4. This method gives overflow indication when a point is repositioned.
5. It is an easy method because each step involves just two additions.

Disadvantage:

1. It involves floating point additions rounding off is done. Accumulations of round off error cause accumulation of error.
2. Rounding off operations and floating point operations consumes a lot of time.
3. It is more suitable for generating line using the software. But it is less suited for hardware implementation.

DDA Algorithm:

Step1: Start Algorithm

Step2: Declare x_1,y_1,x_2,y_2,dx,dy,x,y as integer variables.

Step3: Enter value of x_1,y_1,x_2,y_2.

Step4: Calculate dx = x_2-x_1

Step5: Calculate dy = y_2-y_1

Step6: If ABS (dx) > ABS (dy)
Then step = abs (dx)
Else

Step7: x_{inc}=dx/step
y_{inc}=dy/step
assign x = x_1
assign y = y_1

Step8: Set pixel (x, y)

Step9: x = x + x_{inc}
y = y + y_{inc}
Set pixels (Round (x), Round (y))

Step10: Repeat step 9 until x = x_2

Step11: End Algorithm

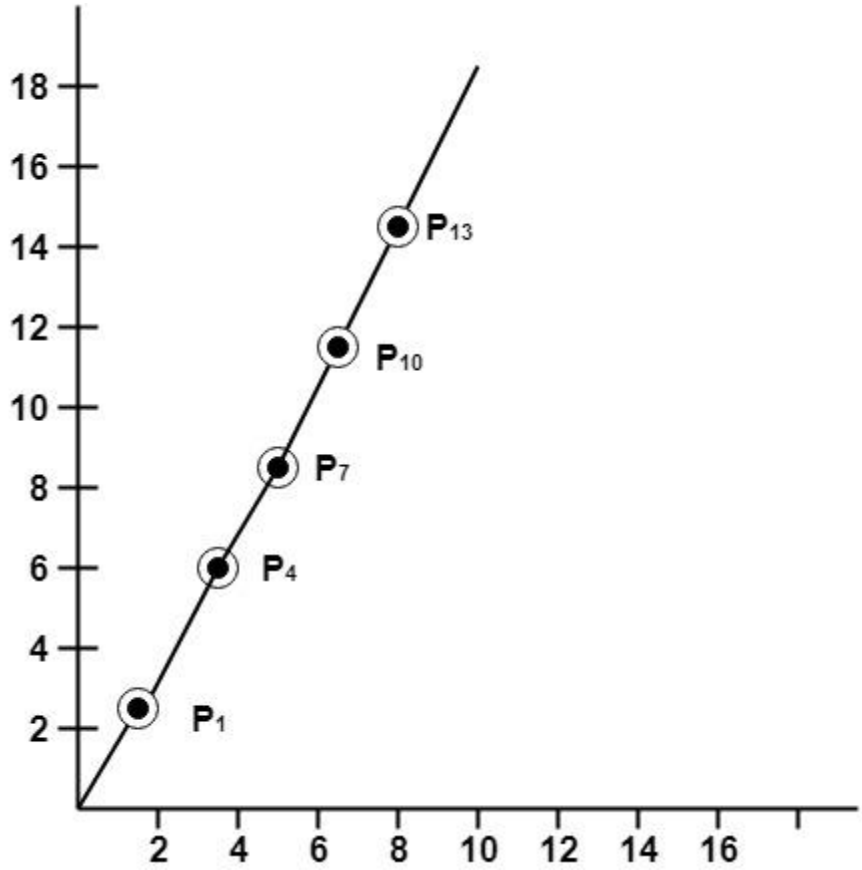
Example: If a line is drawn from (2, 3) to (6, 15) with use of DDA. How many points will needed to generate such line?

Solution: P_1 (2,3) P_{11} (6,15)

x_1=2
y_1=3
x_2= 6
y_2=15
dx = 6 - 2 = 4
dy = 15 - 3 = 12
m = $\frac{dy}{dx} = \frac{12}{4}$

For calculating next value of x takes x = x + $\frac{1}{m}$

$P_1(2, 3)$	point plotted
$P_2(2\frac{1}{3}, 4)$	point plotted
$P_3(2\frac{2}{3}, 5)$	point not plotted
$P_4(3, 6)$	point plotted
$P_5(3\frac{1}{3}, 7)$	point not plotted
$P_6(3\frac{2}{3}, 8)$	point not plotted
$P_7(4, 9)$	point plotted
$P_8(4\frac{1}{3}, 10)$	point not plotted
$P_9(4\frac{2}{3}, 11)$	point not plotted
$P_{10}(5, 12)$	point plotted
$P_{11}(5\frac{1}{3}, 13)$	point not plotted
$P_{12}(5\frac{2}{3}, 14)$	point not plotted
$P_{13}(6, 15)$	point plotted



Program to implement DDA Line Drawing Algorithm:

```
#include<graphics.h>
#include<conio.h>
#include<stdio.h>
void main()
{
    intgd = DETECT ,gm, i;
    float x, y,dx,dy,steps;
    int x0, x1, y0, y1;
    initgraph(&gd, &gm, "C:\\TC\\BGI");
    setbkcolor(WHITE);
    x0 = 100 , y0 = 200, x1 = 500, y1 = 300;
    dx = (float)(x1 - x0);
    dy = (float)(y1 - y0);
    if(dx>=dy)
        {
      steps = dx;
    }
    else
        {
      steps = dy;
    }
    dx = dx/steps;
    dy = dy/steps;
    x = x0;
    y = y0;
    i = 1;
    while(i<= steps)
    {
      putpixel(x, y, RED);
      x += dx;
      y += dy;
      i=i+1;
    }
    getch();
    closegraph();
}
```

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Output:

Bresenham's(Mid-Point) Line Algorithm

This algorithm is used for scan converting a line. It was developed by Bresenham. It is an efficient method because it involves only integer addition, subtractions, and multiplication operations. These operations can be performed very rapidly so lines can be generated quickly.

In this method, next pixel selected is that one who has the least distance from true line.

The method works as follows:

Assume a pixel $P_1'(x_1',y_1')$,then select subsequent pixels as we work our may to the night, one pixel position at a time in the horizontal direction toward $P_2'(x_2',y_2')$.

Once a pixel in choose at any step

The next pixel is

1. Either the one to its right (lower-bound for the line)
2. One top its right and up (upper-bound for the line)

The line is best approximated by those pixels that fall the least distance from the path between P_1',P_2'.

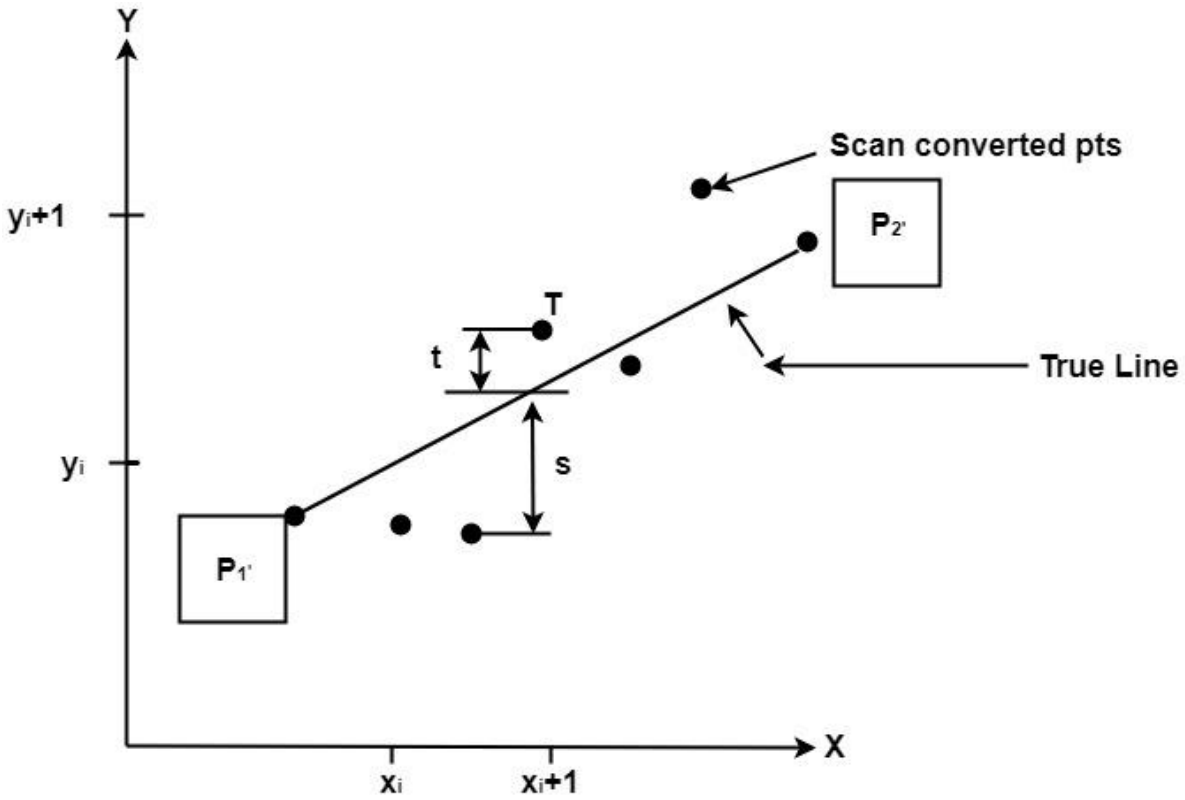


Fig: Scan Converting a line.

To chooses the next one between the bottom pixel S and top pixel T.

If S is chosen

We have $x_{i+1}=x_i+1$ and $y_{i+1}=y_i$

If T is chosen

We have $x_{i+1}=x_i+1$ and $y_{i+1}=y_i+1$

The actual y coordinates of the line at $x = x_{i+1}$ is

$y=mx_{i+1}+b$

$$\boxed{y = m(x_i + 1) + b}$$

The distance from S to the actual line in y direction

$s = y-y_i$

The distance from T to the actual line in y direction

$t = (y_i+1)-y$

Now consider the difference between these 2 distance values

s - t

When (s-t) <0 ⟹ s < t

The closest pixel is S

When (s-t) ≥0 ⟹ s < t

The closest pixel is T

This difference is

$$s-t = (y-yi)-[(yi+1)-y]$$
$$= 2y - 2yi -1$$

$$\boxed{s - t = 2m(x_i + 1) + 2b - 2y_i - 1}$$ [Putting the value of (1)]

Substituting m by $\frac{\triangle y}{\triangle x}$ and introducing decision variable

$$d_i=\triangle x\ (s-t)$$
$$d_i=\triangle x\ (2\ \frac{\triangle y}{\triangle x}\ (x_i+1)+2b-2y_i-1)$$
$$=2\triangle xy_i-2\triangle y-1\triangle x.2b-2y_i\triangle x-\triangle x$$
$$d_i=2\triangle y.x_i-2\triangle x.y_i+c$$

Where c= 2△y+△x (2b-1)

We can write the decision variable d_{i+1} for the next slip on

$$d_{i+1}=2\triangle y.x_{i+1}-2\triangle x.y_{i+1}+c$$
$$d_{i+1}-d_i=2\triangle y.(x_{i+1}-x_i)- 2\triangle x(y_{i+1}-y_i)$$

Since x_(i+1)=x_i+1,we have

$$d_{i+1}+d_i=2\triangle y.(x_i+1-x_i)- 2\triangle x(y_{i+1}-y_i)$$

Special Cases

If chosen pixel is at the top pixel T (i.e., $d_i \geq 0$)⟹ $y_{i+1}=y_i+1$

$$d_{i+1}=d_i+2\triangle y-2\triangle x$$

If chosen pixel is at the bottom pixel T (i.e., $d_i<0$)⟹ $y_{i+1}=y_i$

$$d_{i+1}=d_i+2\triangle y$$

Finally, we calculate d_1

$$d_1=\triangle x[2m(x_1+1)+2b-2y_1-1]$$
$$d_1=\triangle x[2(mx_1+b-y_1)+2m-1]$$

Since $mx_1+b-y_i=0$ and $m = \frac{\Delta y}{\Delta x}$, we have

$d_1=2\Delta y-\Delta x$

Advantage:

1. It involves only integer arithmetic, so it is simple.

2. It avoids the generation of duplicate points.

3. It can be implemented using hardware because it does not use multiplication and division.

4. It is faster as compared to DDA (Digital Differential Analyzer) because it does not involve floating point calculations like DDA Algorithm.

Disadvantage:

1. This algorithm is meant for basic line drawing only Initializing is not a part of Bresenham's line algorithm. So to draw smooth lines, you should want to look into a different algorithm.

Bresenham's Line Algorithm:

Step1: Start Algorithm

Step2: Declare variable $x_1,x_2,y_1,y_2,d,i_1,i_2,dx,dy$

Step3: Enter value of x_1,y_1,x_2,y_2

Where $x_1,y_1$are coordinates of starting point

And x_2,y_2 are coordinates of Ending point

Step4: Calculate $dx = x_2-x_1$

Calculate $dy = y_2-y_1$

Calculate $i_1=2*dy$

Calculate $i_2=2*(dy-dx)$

Calculate $d=i_1-dx$

Step5: Consider (x, y) as starting point and x_{end}as maximum possible value of x.

If $dx < 0$

Then $x = x_2$

$y = y_2$

$x_{end}=x_1$

If $dx > 0$

Then $x = x_1$
$y = y_1$
$x_{end}=x_2$

Step6: Generate point at (x,y)coordinates.

Step7: Check if whole line is generated.
If $x >= x_{end}$
Stop.

Step8: Calculate co-ordinates of the next pixel
If $d < 0$
Then $d = d + i_1$
If $d \geq 0$
Then $d = d + i_2$
Increment $y = y + 1$

Step9: Increment $x = x + 1$

Step10: Draw a point of latest (x, y) coordinates

Step11: Go to step 7

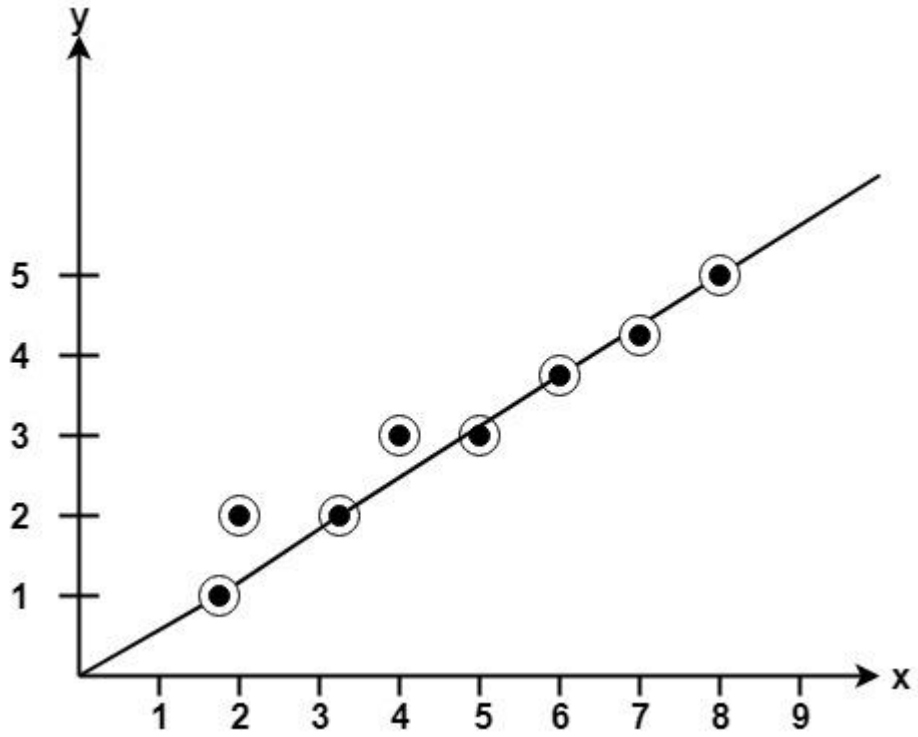
Step12: End of Algorithm

Example: Starting and Ending position of the line are (1, 1) and (8, 5). Find intermediate points.

Solution: $x_1=1$
$y_1=1$
$x_2=8$
$y_2=5$
$dx= x_2-x_1=8-1=7$
$dy=y_2-y_1=5-1=4$
$I_1=2* \Delta y=2*4=8$
$I_2=2*(\Delta y-\Delta x)=2*(4-7)=-6$
$d = I_1-\Delta x=8-7=1$

x	y	$d=d+I_1$ or I_2
1	1	$d+I_2=1+(-6)=-5$

2	2	$d+I_1=-5+8=3$
3	2	$d+I_2=3+(-6)=-3$
4	3	$d+I_1=-3+8=5$
5	3	$d+I_2=5+(-6)=-1$
6	4	$d+I_1=-1+8=7$
7	4	$d+I_2=7+(-6)=1$
8	5	



Program to implement Bresenham's Line Drawing Algorithm:

```
1. #include<stdio.h>
2. #include<graphics.h>
3. void drawline(int x0, int y0, int x1, int y1)
4. {
5.     int dx, dy, p, x, y;
6.     dx=x1-x0;
```

```
    dy=y1-y0;
    x=x0;
    y=y0;
    p=2*dy-dx;
    while(x<x1)
    {
        if(p>=0)
        {
            putpixel(x,y,7);
            y=y+1;
            p=p+2*dy-2*dx;
        }
        else
        {
            putpixel(x,y,7);
            p=p+2*dy;}
            x=x+1;
        }
}
int main()
{
    int gdriver=DETECT, gmode, error, x0, y0, x1, y1;
    initgraph(&gdriver, &gmode, "c:\\turboc3\\bgi");
    printf("Enter co-ordinates of first point: ");
    scanf("%d%d", &x0, &y0);
    printf("Enter co-ordinates of second point: ");
    scanf("%d%d", &x1, &y1);
    drawline(x0, y0, x1, y1);
    return 0;
}
```

Output:

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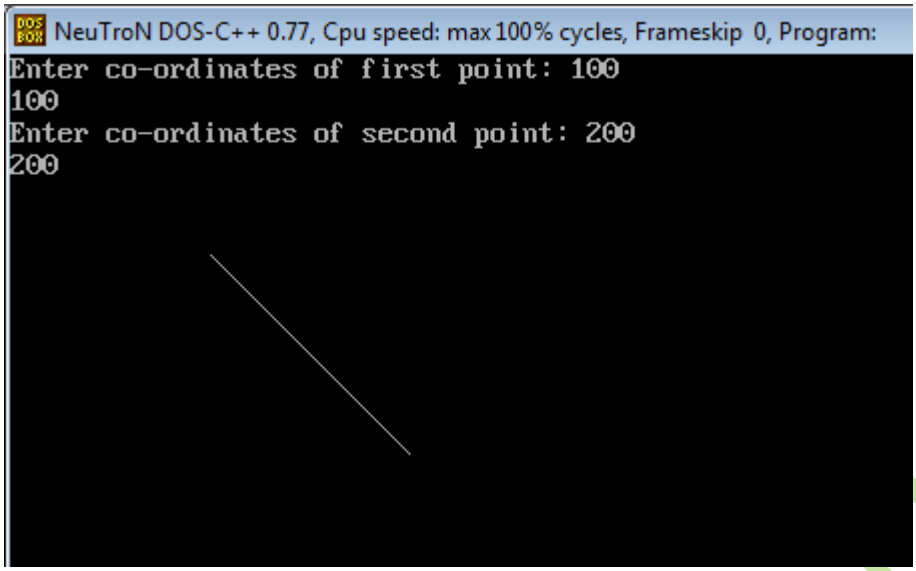


Differentiate between DDA Algorithm and Bresenham's Line Algorithm:

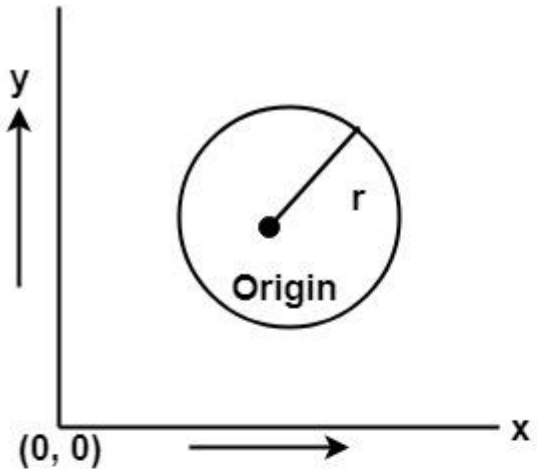
DDA Algorithm	Bresenham's Line Algorithm
1. DDA Algorithm use floating point, i.e., Real Arithmetic.	1. Bresenham's Line Algorithm use fixed point, i.e., Integer Arithmetic
2. DDA Algorithms uses multiplication & division its operation	2.Bresenham's Line Algorithm uses only subtraction and addition its operation
3. DDA Algorithm is slowly than Bresenham's Line Algorithm in line drawing because it uses real arithmetic (Floating Point operation)	3. Bresenham's Algorithm is faster than DDA Algorithm in line because it involves only addition & subtraction in its calculation and uses only integer arithmetic.
4. DDA Algorithm is not accurate and efficient as Bresenham's Line Algorithm.	4. Bresenham's Line Algorithm is more accurate and efficient at DDA Algorithm.
5.DDA Algorithm can draw circle and curves but are not accurate as Bresenham's Line Algorithm	5. Bresenham's Line Algorithm can draw circle and curves with more accurate than DDA Algorithm.

Defining a Circle:

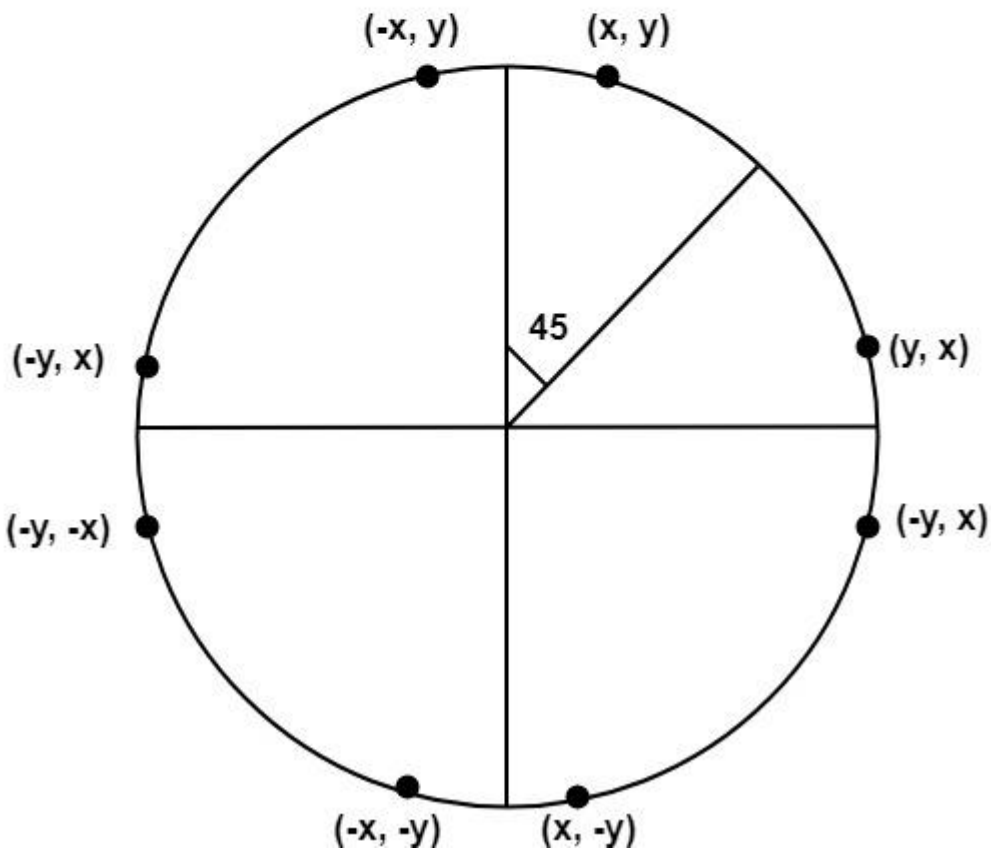
Circle is an eight-way symmetric figure. The shape of circle is the same in all quadrants. In each quadrant, there are two octants. If the calculation of the point of one octant is

done, then the other seven points can be calculated easily by using the concept of eight-way symmetry.

For drawing, circle considers it at the origin. If a point is $P_1(x, y)$, then the other seven points will be



P2 (x, -y)
P3(-x, -y)
P4 (-x, y)
P5 (y, x)
P6 (y, -x)
P7 (-y, -x)
P8 (-y, x)



So we will calculate only 45°arc. From which the whole circle can be determined easily.

If we want to display circle on screen then the putpixel function is used for eight points as shown below:

putpixel (x, y, color)
putpixel (x, -y, color)
putpixel (-x, y, color)
putpixel (-x, -y, color)
putpixel (y, x, color)
putpixel (y, -x, color)
putpixel (-y, x, color)
putpixel (-y, -x, color)

Example: Let we determine a point (2, 7) of the circle then other points will be (2, -7), (-2, -7), (-2, 7), (7, 2), (-7, 2), (-7, -2), (7, -2)

These seven points are calculated by using the property of reflection. The reflection is accomplished in the following way:

The reflection is accomplished by reversing x, y co-ordinates.

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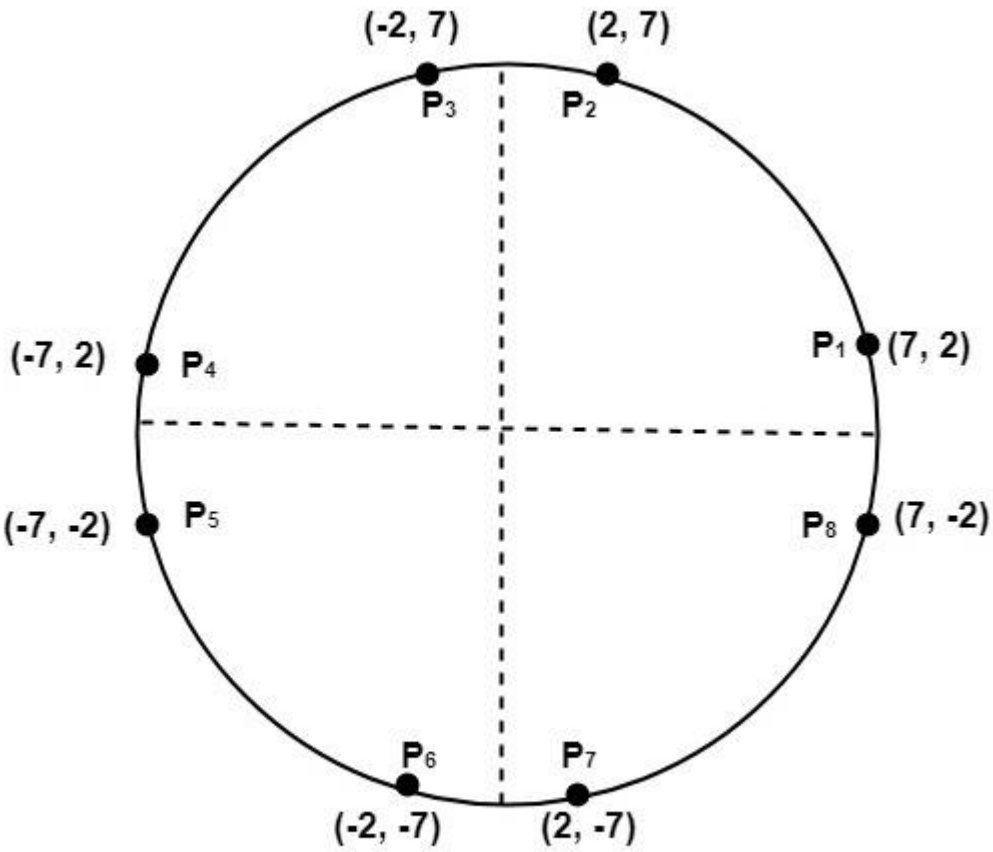

Eight way symmetry of a Circle

There are two standards methods of mathematically defining a circle centered at the origin.

1. Defining a circle using Polynomial Method
2. Defining a circle using Polar Co-ordinates